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ME 470

Introduction to Aerodynamics

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Semester Project 1 Report

FLOWY: A 2D POTENTIAL FLOW PROGRAM

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1. INTRODUCTION

Flowy is a program for two-dimensional, incompressible and irrotational (potential) flows. Elementary flows (uniform flow, source or sink, point vortex, doublet and sector flow) and straight symmetry walls are added to a scene, and the program superposes them. The streamlines with direction arrows are redrawn after every change. Stagnation points, equipotential lines, velocity and pressure maps, solid bodies and the pressure on their surfaces can also be shown.

A short interactive guide is built into the program. On the first start, a pixel-art beaver on the menu bar offers a tour of about two minutes, in which a cylinder flow is built step by step. The beaver can be called at any time by clicking him or by pressing P. A reader who prefers to start at once can open the program and follow the tour. For more detailed information on every function, reading this guide is recommended.

In this guide, names of buttons, menus and fields are written in bold, and a path through the menus is written as **Menu → Item**. Instructions are given as numbered steps.

2. INSTALLATION AND START

2.1 Requirements

A computer with 64-bit Windows 10 or 11 is required. Nothing else has to be installed: Python, the libraries and all other files are included in the program folder. No internet connection is needed. The program is delivered in two forms with the same content: a portable folder and an installer.

2.2 Portable Version

1. Right-click the zip file and select **Extract All…**, then **Extract**. The program does not run from inside a zip file.
2. Open the folder **3_Program\Flowy 1.0.0**.
3. Double-click **Flowy.exe**.
4. If Windows shows “Windows protected your PC”, click **More info**, then **Run anyway**. This message appears because the program is not signed.

2.3 Installer Version

1. Extract the zip file and double-click **3_Program\Flowy-v1.0.0-setup.exe** (with **More info → Run anyway** if Windows asks).

2. Select the language and whether the program is installed only for the current user (no administrator rights needed) or for all users.
3. Select the installation folder and the Start menu folder.
4. Select the options: a desktop shortcut, opening .pflow scene files with Flowy, and starting Flowy with Windows.
5. Click **Install**, then **Finish**. The program is started from the Start menu or from the desktop.

To uninstall, select **Settings → Apps → Installed apps → Flowy → Uninstall**. The uninstaller asks whether the settings and data are also deleted; saved scene files (.pflow) are never deleted.

2.4 First Start

The first start takes 15–30 s, because the fast calculation routines are prepared for the processor once; a loading screen shows the progress. Later starts take a few seconds. The program starts with an empty plot.

2.5 Running from the Source Code

Running from the source code is optional and is not needed for the program above.

1. Install Python 3.13 from python.org and select “Add python.exe to PATH” during the installation.
2. Open a terminal in the folder **4_Source_Code** (in File Explorer: click the address bar, type cmd and press Enter).
3. Type python -m pip install -r requirements.txt and press Enter (once; an internet connection is needed).
4. Type python main.py and press Enter.

3. THE MAIN WINDOW

The parts of the main window are shown in Figure 1 and explained in Table 1.

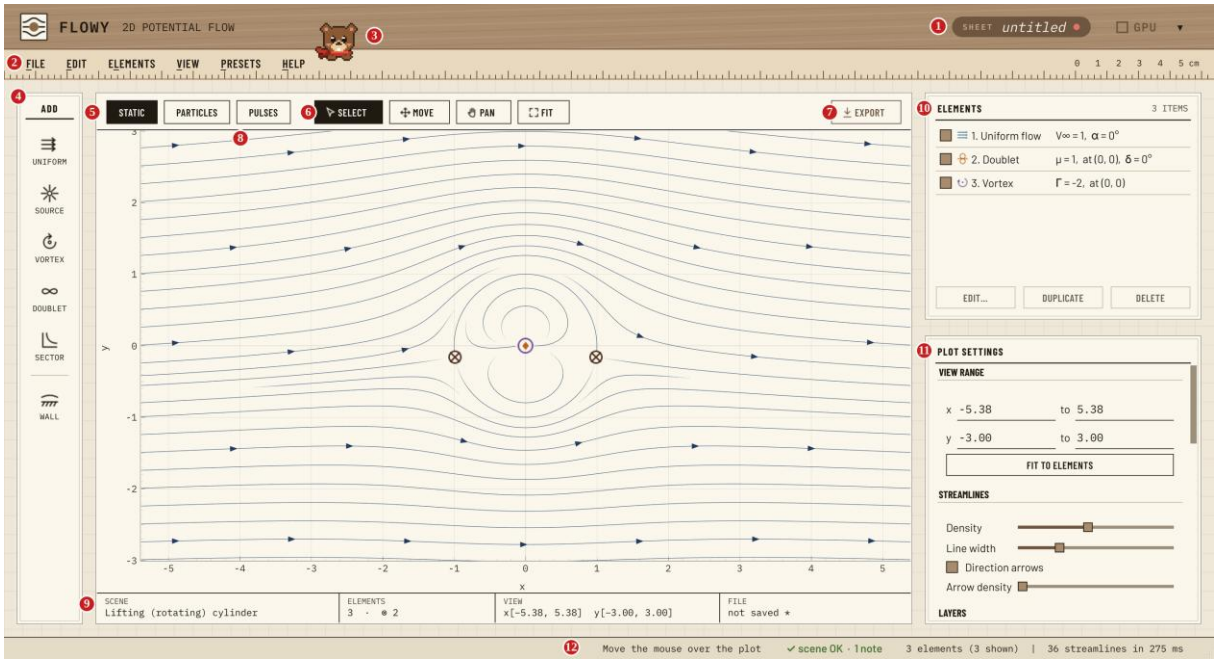


Figure 1. Main window with the lifting cylinder scene; the numbers refer to Table 1

Table 1. Parts of the main window

No.	Part	Function
1	Header	Program name, file name of the scene (SHEET; a red dot means unsaved changes), GPU box
2	Menus	File, Edit, Elements, View, Presets and Help
3	Beaver	Built-in guide: menu, questions and answers, tour (Section 9.1)
4	Add palette	One button per element type; the name is shown on hover
5	View mode	Static (streamlines with arrows), Particles and Pulses (animation)
6	Mouse mode	Select, Move and Pan; Fit shows all elements
7	Export	Figure, animation or data export
8	Plot	Streamlines, stagnation points and the other layers, with labeled x and y axes
9	Information bar	Scene title, number of elements, view range and file state
10	Elements panel	List of all elements; Edit, Duplicate, Delete; tick boxes to hide elements
11	Plot settings	View range, streamline density, arrows, layers, animation and computation
12	Status bar	Values under the mouse and the scene check (OK, warnings or errors)

4. ADDING AND EDITING ELEMENTS

4.1 Adding an Element

1. Click a button in the Add palette (or select **Elements → Add**). The element is placed in the middle of the view. Alternatively, drag the button onto the plot to place the element at that point.

2. Enter the values in the parameter dialog (Figure 2). **Tab** moves to the next field. The complex potential of the element is shown at the top of the dialog.
3. Click **Apply** to add the element, or **Cancel** to add nothing.

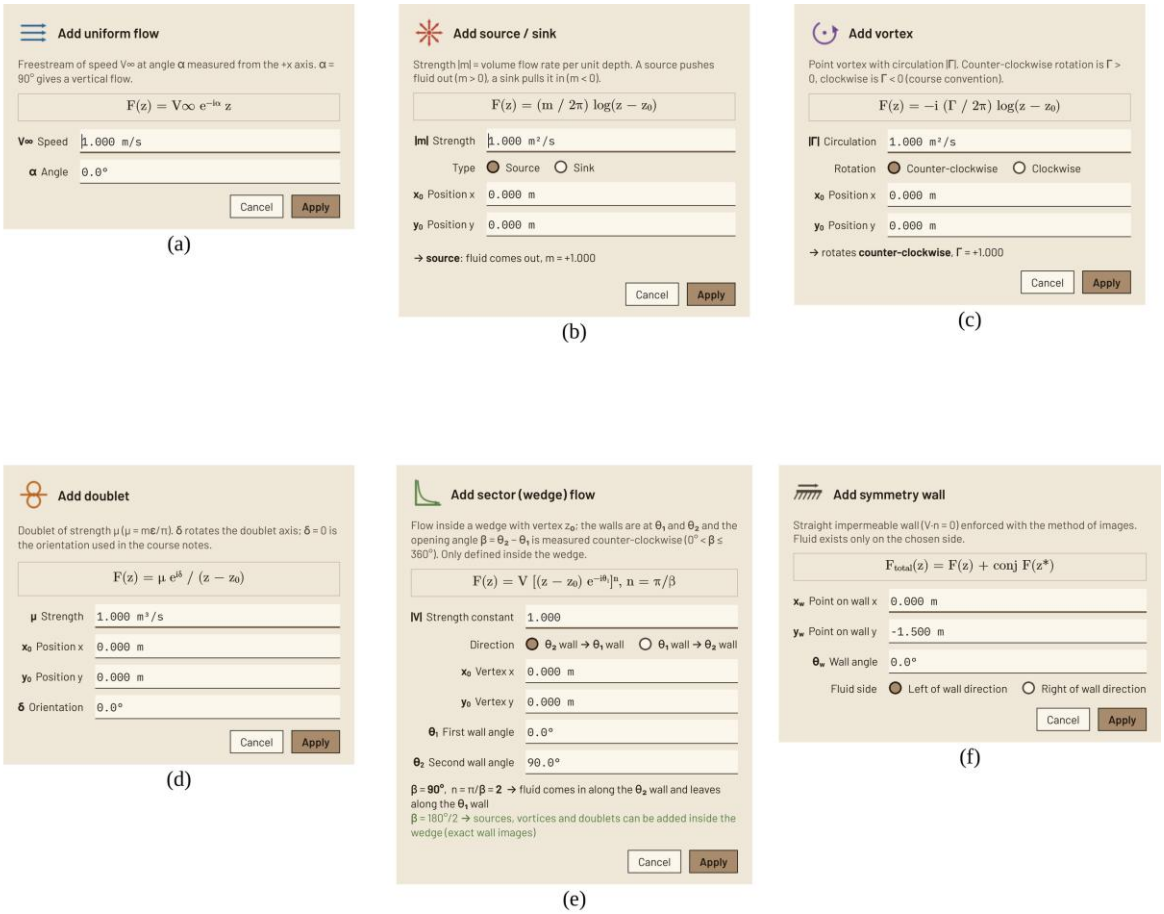


Figure 2. Parameter dialogs: (a) uniform flow, (b) source / sink, (c) point vortex, (d) doublet, (e) sector flow, (f) symmetry wall

The inputs of each element are given in Table 2. Lengths are in metres and angles in degrees, measured counter-clockwise from the $+x$ axis. The circulation Γ is positive for counter-clockwise rotation; this is opposite to the sign used in [1].

Table 2. Inputs of the elements

Element	Inputs	Meaning
Uniform flow	V_∞ [m/s], α [°]	Speed and direction of the free stream
Source / sink	$\lvert m\rvert$ [m^2/s], type, x_0, y_0	Volume flow rate per unit depth; a source pushes fluid out ($m > 0$), a sink pulls it in ($m < 0$)
Point vortex	$\lvert\Gamma\rvert$ [m^2/s], rotation, x_0, y_0	Circulation; counter-clockwise is $\Gamma > 0$
Doublet	μ [m^3/s], x_0, y_0, δ [°]	Strength and axis angle; with a uniform flow and $\delta = \alpha$ it forms a cylinder of radius $a = \sqrt{\mu/V_\infty}$
Sector flow	$\lvert V\rvert$, direction, vertex x_0, y_0, θ_1, θ_2 [°]	Flow inside the wedge between the walls at θ_1 and θ_2; the direction sets which wall the fluid comes in along
Symmetry wall	point x_w, y_w, angle θ_w [°], fluid side	Straight impermeable wall; the fluid is only on the selected side

4.2 Rules for Walls and Sector Flow

Every flow element gets mirror images in every wall, so that no flow passes through the wall. One wall, two walls that meet at $180°/k$ (for example 90°, 60°, 45°) and two parallel walls (a channel) are exact. A uniform flow must be parallel to every wall. A sector flow exists only inside its wedge; its two sides act as walls for the other elements. Other elements must be placed inside the wedge, and they can be added only when $\beta = \theta_2 - \theta_1 = 180°/k$. Only one sector flow is allowed, and it cannot be combined with symmetry walls. Scenes that break these rules are reported as errors.

4.3 Warnings and Errors

The values are checked while they are typed, and the messages are shown in the dialog (Figure 3). A warning (⚠) means that the result may not be the expected one, for example a doublet whose axis is not along the flow. An error (✕) means that the scene is physically inconsistent, for example a source behind a wall. In that case the button changes to **Apply anyway**; the element can still be added, and a red strip is shown on the plot until the problem is fixed. The scene check in the status bar shows the same messages for the whole scene; click it to see the list. Two singular points at the same place (for example a doublet and a vortex at the center of a lifting cylinder) are allowed, and only a note is shown.

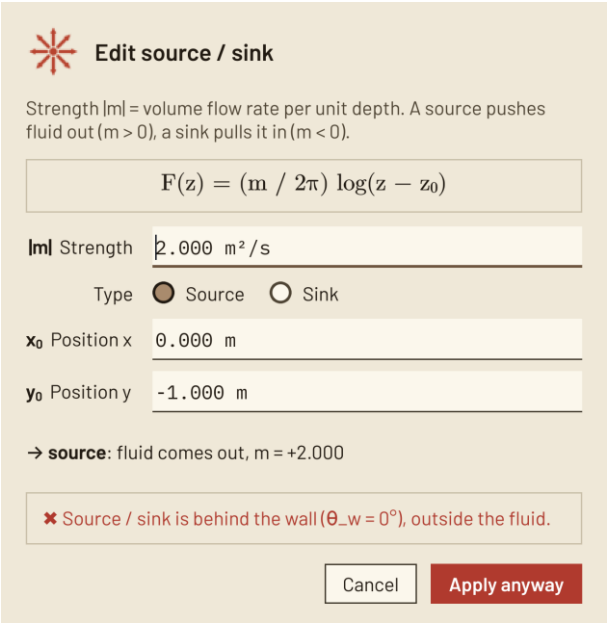
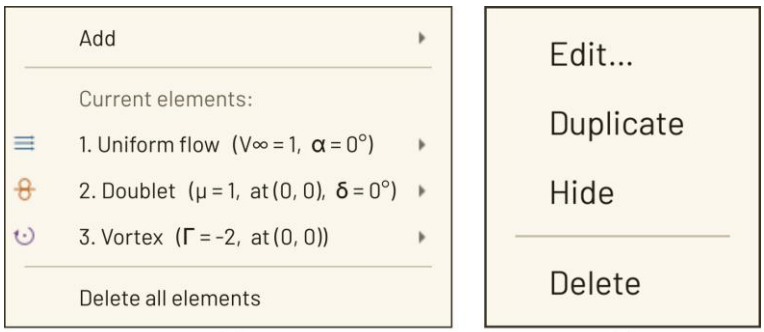


Figure 3. Parameter dialog of a source placed behind a wall: the error message and the Apply anyway button

4.4 Editing, Moving and Deleting

1. **Edit**: double-click the element in the Elements panel or its symbol on the plot, or select it and press Enter, or use **Elements → (element) → Edit…** (Figure 4).
2. **Move**: select **Move** above the plot (or press M) and drag a symbol or a wall. Hold Shift to snap to the grid; press Esc to cancel. The streamlines follow during the drag.
3. **Hide** an element temporarily: clear its tick box in the Elements panel.
4. **Duplicate**: click **Duplicate** or press Ctrl+D; the copy is shifted by $(0.5, 0.5)$.
5. **Delete**: click **Delete** or press the Delete key. **File → Clear all** removes all elements.
6. **Undo / Redo**: press Ctrl+Z / Ctrl+Y. Every change can be undone, also loading a preset.



(a) (b)

Figure 4. (a) Elements menu with the current elements and their parameters, (b) menu of one element

5. THE PLOT

5.1 Navigation

Zoom with the mouse wheel; the point under the mouse stays in place. Pan by dragging an empty part of the plot or in the Pan mode (H). **Fit** (F) shows all elements. Exact plot limits are typed in **Plot settings → View range**. The status bar shows the coordinates under the mouse together with u, v, $|V|$, C_p, ϕ and ψ there. Stagnation points are marked with ⊗; click one to see its coordinates and the dividing streamlines through it.

5.2 Plot Settings and Layers

The **Plot settings** panel (Figure 5) contains the view range, the streamline density, the line width, the direction arrows and their density, the layers, the animation settings and the computation settings. The plot settings are saved with the scene. The layers are given in Table 3, and four of them are shown in Figure 6. The layers can also be switched in **View → Layers**.

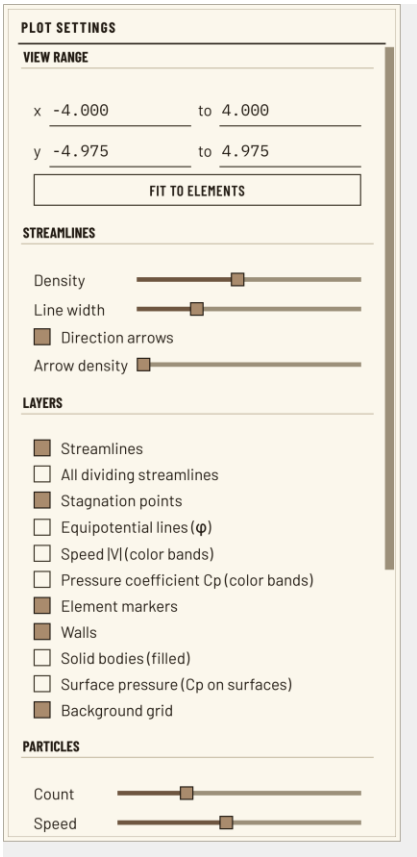


Figure 5. Plot settings panel: view range, streamlines and layers

Table 3. Layers of the plot

Layer	What is drawn
Streamlines	Evenly spaced streamlines with direction arrows
All dividing streamlines	The streamlines through every stagnation point (thick lines)
Stagnation points	Points with $V = 0$ ($\otimes$)
Equipotential lines	Lines of constant ϕ, perpendicular to the streamlines
Speed $\lvert V \rvert$	Color bands of the velocity magnitude
Pressure coefficient C_p	Color bands of $C_p = 1 - \left(V/V_{ref}\right)^2$; V_{ref} is V_∞, or a value set in Computation
Element markers, walls, grid	Symbols of the elements, the walls and the background grid
Solid bodies	Regions that the fluid never enters (cylinder, oval, half-body), filled
Surface pressure	C_p on body surfaces and walls; opens the Surface pressure panel (Section 5.4)

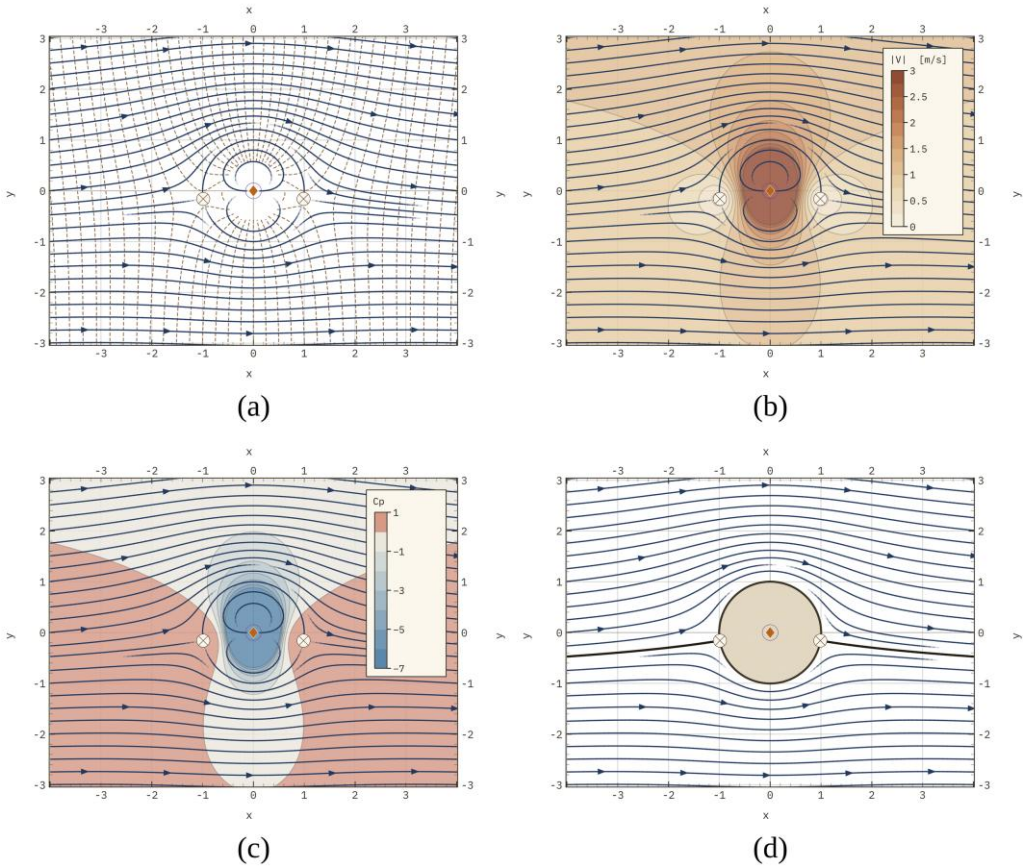


Figure 6. Layers on the lifting cylinder: (a) equipotential lines, (b) speed $|V|$, (c) pressure coefficient C_p, (d) dividing streamlines and solid body

5.3 Animation

The buttons at the left above the plot select the view mode. **Static** shows the streamlines with arrows. **Particles** shows tracer particles moving with the flow; their color is the speed. **Pulses** shows pulses running along the streamlines at the local speed (Figure 7). Press Space to pause and continue. The number, speed and trail length are set in the plot settings.

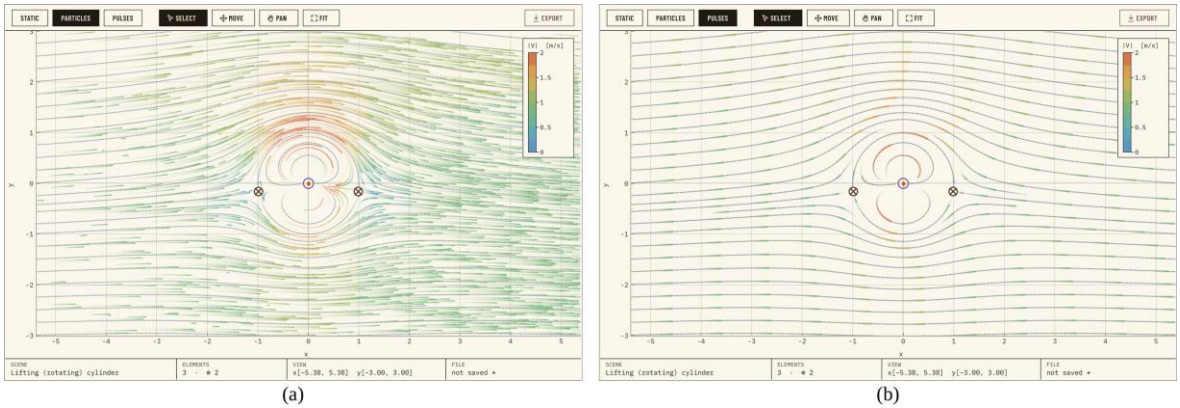


Figure 7. Animation of the lifting cylinder: (a) Particles, (b) Pulses

5.4 Solid Bodies and Surface Pressure

Select **View → Layers → Surface pressure**. The surfaces of the bodies and the walls are colored with C_p, and the **Surface pressure** panel opens on the left (Figure 8). The panel shows the C_p distribution along the surface; for a cylinder, the analytic curve is drawn dashed for comparison. For a closed body, the lift and drag per unit span from the pressure integral are shown next to the Kutta–Joukowski lift and the zero drag of d'Alembert's paradox. **Export CSV…** saves the values.

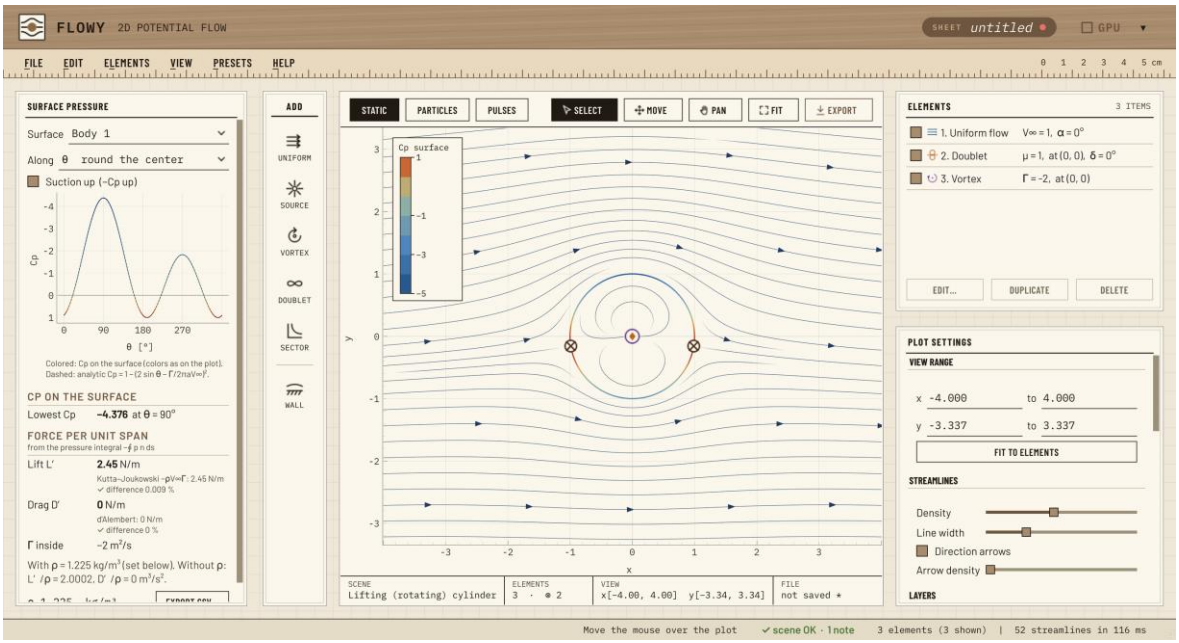


Figure 8. Surface pressure panel (left) with the C_p distribution, lift and drag of the lifting cylinder

6. FILES, PRESETS AND EXPORT

6.1 Files

The **File** menu (Figure 9(a)) contains **New** (Ctrl+N), **Open…** (Ctrl+O), **Open recent**, **Save** (Ctrl+S), **Save as…** (Ctrl+Shift+S), the export functions, **Clear all** and **Exit**. A scene file (.pflow) stores the element types, all parameters, the walls, the title, the view and the plot settings. When a scene is saved and opened again, the element list and the plot are the same. Unsaved changes are marked in the header, and the program asks before they could be lost. If the program closes unexpectedly, the unsaved scene is offered for restoring at the next start.

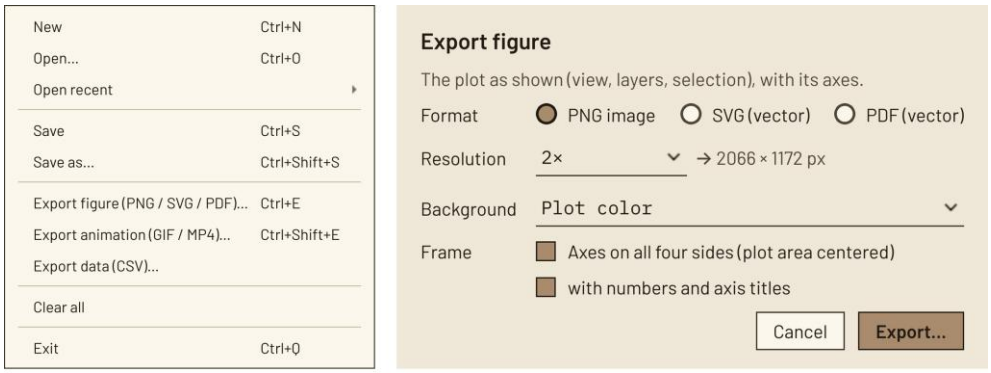
6.2 Presets

Ready-made scenes are in the **Presets** menu. The first group, **Verification cases (assignment)**, contains the verification cases of the assignment (Section 7.2). Further groups contain bodies, sector flows and walls. A preset is loaded as a new, unsaved scene.

6.3 Export

Select **Export** above the plot or the **File** menu (Figure 9(b)):

1. **Figure** (Ctrl+E): PNG, SVG or PDF of the plot as shown, with the axes; resolution 1–4 times the screen.
2. **Animation** (Ctrl+Shift+E): GIF or MP4 of the Particles or Pulses mode.
3. **Data** (CSV): the elements, the stagnation points, the streamline points, the field on a grid (u, v, $|V|$, C_p, ϕ, ψ) and the surface pressure.



(a) (b)

Figure 9. (a) File menu, (b) Export figure dialog

7. VERIFICATION

7.1 Running the Verification

Select **Help → Run verification…**. The program checks 108 results against analytic solutions in about 15 s (Figure 10). The checks are grouped as the analytic formulas, the verification scenes of the assignment and the numerical methods. Select a row to see the theory, the computation and the tolerance of the check. **Open full report** opens the complete report with figures in the web browser.

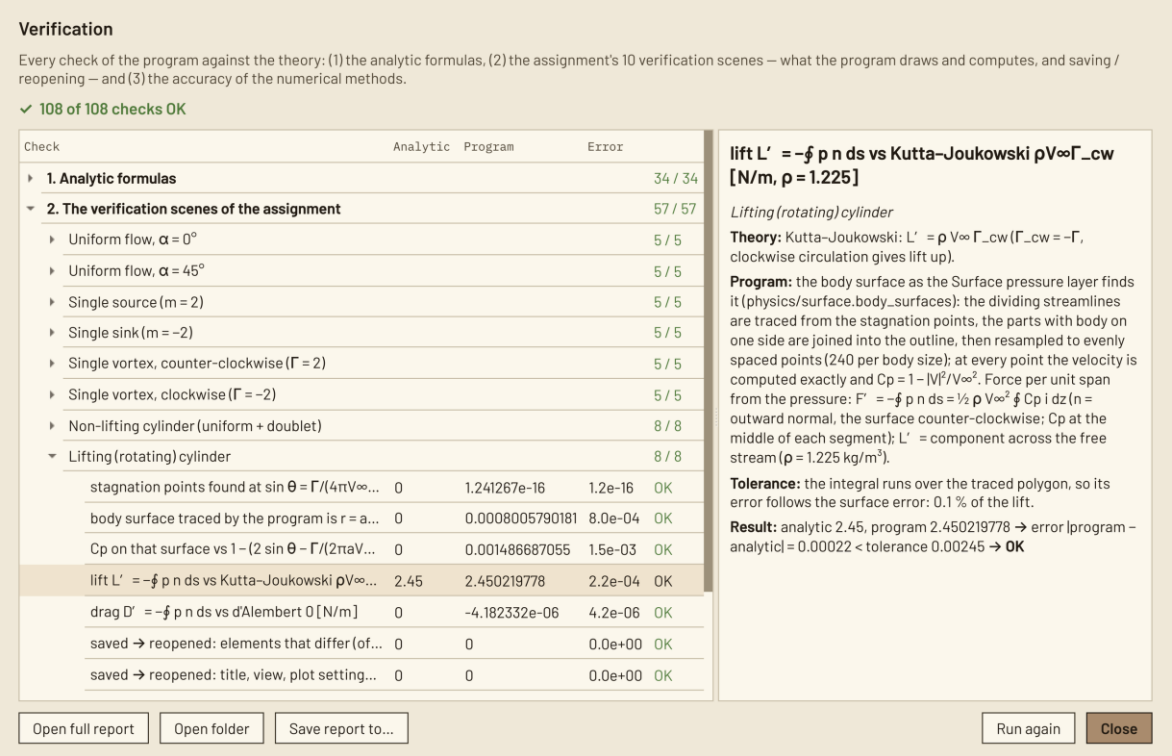


Figure 10. Verification window: the results of all checks and the explanation of the selected check

7.2 Repeating the Verification Cases of the Assignment

Each case of the assignment can be checked by hand with the presets in Table 4.

Table 4. Verification cases of the assignment in the program

Case	Preset (Presets → Verification cases)	What to check
1. Uniform flow	Uniform flow, $\alpha = 0°$; Uniform flow, $\alpha = 45°$	Straight streamlines at α; arrows in the flow direction
2. Source, sink, vortex	Single source; Single sink; Single vortex (ccw, cw)	Radial lines out of the source and into the sink; circles turning in the sense of Γ
3. Cylinder	Non-lifting cylinder; Lifting (rotating) cylinder	Stagnation points on $r = a$; Surface pressure: C_p and lift against Kutta–Joukowski
4. Symmetry wall	Symmetry wall: source near a wall	No streamline crosses the wall; the verification report gives the largest $\lvert V \cdot n \rvert$ on it
5. Save and reopen	Any scene	**File → Save as…**, **File → New**, **File → Open…**: the same element list and plot

8. WORKED EXAMPLES

8.1 Lifting Cylinder

1. Select **File → New**.
2. Click **Uniform** in the palette, keep $V_\infty = 1$ m/s and $\alpha = 0°$, and click **Apply** (Figure 11(a)).

3. Click **Doublet**, set $\mu = 1\ \mathrm{m}^3/\mathrm{s}$ at $(0,0)$ with $\delta = 0°$, and click **Apply**. The flow over a cylinder of radius $a = 1\ \mathrm{m}$ appears (Figure 11(b)).
4. Click **Vortex**, set $|\Gamma| = 2\ \mathrm{m}^2/\mathrm{s}$, select **Clockwise**, keep $(0,0)$, and click **Apply**. The stagnation points move below the horizontal axis (Figure 11(c)).
5. Select **View** → **Layers** → **Surface pressure** to see the lift $L' = 2.45\ \mathrm{N/m}$ for $\rho = 1.225\ \mathrm{kg/m}^3$.
6. Press **Particles** to watch the flow, and save the scene with **File** → **Save as…**.

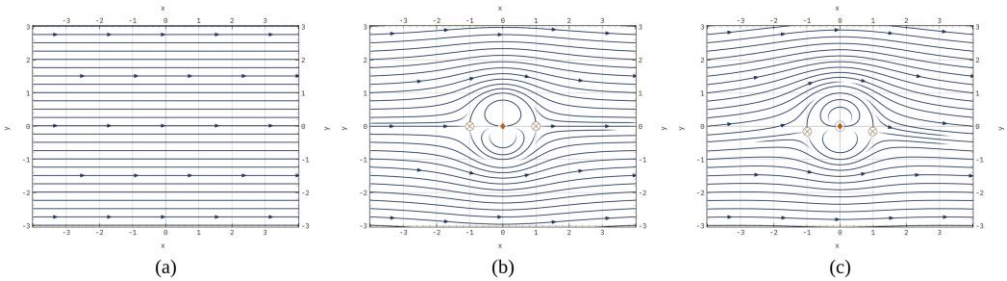


Figure 11. Lifting cylinder step by step: (a) uniform flow, (b) uniform flow and doublet, (c) with a clockwise vortex, $\Gamma = -2$

8.2 Source near a Wall

1. Select **File** → **New** and click **Wall**. Set the point $(0,0)$, $\theta_w = 0°$ and the fluid side **Left of wall direction** (the fluid is above the x axis), and click **Apply**.
2. Click **Source**, set $|m| = 2\ \mathrm{m}^2/\mathrm{s}$ at $(0,1)$, and click **Apply**. The wall is a streamline, and a stagnation point appears on the wall under the source (Figure 12(a)).

8.3 Source in a Sector Flow

1. Select **File** → **New** and click **Sector**. Set $|V| = 1$, the vertex $(0,0)$, $\theta_1 = 0°$ and $\theta_2 = 90°$, and click **Apply**: the flow in a 90° corner.
2. Click **Source**, set $|m| = 2\ \mathrm{m}^2/\mathrm{s}$ at $(1,1)$ inside the wedge, and click **Apply**. The wedge sides remain streamlines; select **View** → **Layers** → **Solid bodies** to see the body formed by the source (Figure 12(b)).

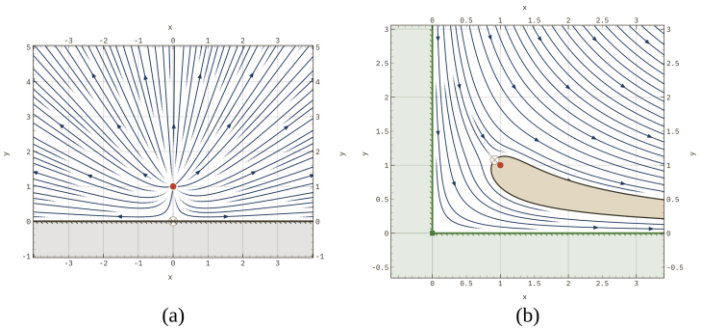
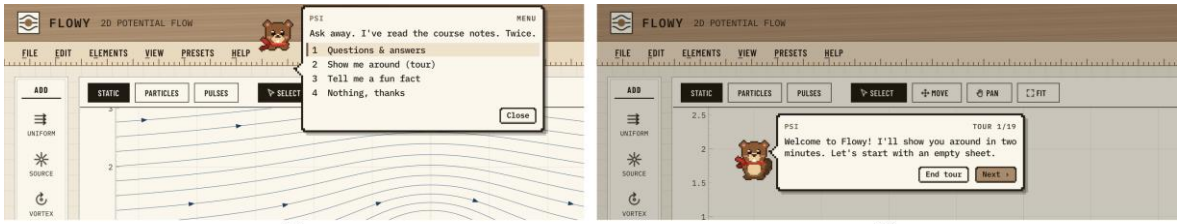


Figure 12. (a) Source $m = 2$ at $(0,1)$ above a wall on the x axis, (b) source $m = 2$ at $(1,1)$ in a 90° sector flow, with its solid body

9. OTHER FEATURES

9.1 The Beaver

The beaver on the menu bar is the built-in guide. Click him or press P to open his menu (Figure 13(a)): questions and answers, the tour and a fun fact. The tour (**Help → Beaver → Show me around**) shows every part of the window and lets the user build a cylinder flow (Figure 13(b)); Enter, Space or → go on, ← goes back and Esc ends it. The beaver can also be dragged onto the plot, where he moves with the flow in the animation modes. He never appears in exports. **View → Beaver** switches him off.



(a) (b)

Figure 13. The beaver: (a) his menu, (b) the first step of the tour

9.2 Graphics Card

The **GPU** box in the header lets the graphics card draw the plot, which makes the animations and busy scenes smoother. Changing it restarts the program in a few seconds; the scene and all settings come back unchanged, also unsaved changes. With the graphics card, the sliders in the plot settings reach higher values. If the card cannot draw, the program returns to drawing with the processor by itself. Exports are the same with and without the card.

9.3 Further Settings

A dark theme is available in **View → Theme**. The panels can be moved by dragging their titles; **View → Panels → Reset layout** restores the original layout. **Help → Sign conventions** shows the signs and angles of all elements, and **Help → User guide** (F1) opens a short version of this guide inside the program.

10. KEYBOARD SHORTCUTS

The default keyboard shortcuts are given in Table 5. They can be changed in **Edit → Keyboard shortcuts…**.

Table 5. Keyboard shortcuts

Key	Action	Key	Action
Ctrl+N	New scene	V	Select mode
Ctrl+O	Open	M	Move mode
Ctrl+S	Save	H	Pan mode
Ctrl+Shift+S	Save as	F	Fit view
Ctrl+E	Export figure	Space	Pause / continue the animation
Ctrl+Shift+E	Export animation	Delete	Delete the selected element
Ctrl+Z	Undo	Enter	Edit the selected element
Ctrl+Y	Redo	Esc	Clear the selection / cancel a drag
Ctrl+D	Duplicate	Shift + drag	Snap to the grid (Move mode)
Ctrl+Q	Exit	F1 / P	User guide / the beaver's menu

11. TROUBLESHOOTING

Common situations and their solutions are given in Table 6.

Table 6. Troubleshooting

Situation	Solution
“Windows protected your PC”	Click **More info → Run anyway** (the program is not signed)
The first start is slow	Normal once (15–30 s); later starts take a few seconds
Nothing is visible on the plot	Press **Fit** (F); check the tick boxes in the Elements panel and the layers
An element shows an error	Read the message in the dialog or click the scene check in the status bar
Panels are moved or closed	Select **View → Panels → Reset layout**
An error window appears	The program and the scene are kept; the log file is in **Help → Open log folder**
The program closed unexpectedly	Start it again; the unsaved scene is offered for restoring

12. CONCLUSION

This guide describes the installation of Flowy, the main window, the adding and editing of elements, the plot and its layers, the files and exports, the verification and three worked examples. With these steps, every flow required by the assignment can be built, checked and saved. The built-in tour and the questions and answers of the beaver give a quick start, and this guide gives the details.

REFERENCES

[1] Anderson, J. D., Jr., 2017, Fundamentals of Aerodynamics, 6th ed., McGraw-Hill Education, New York.